

Capturing traffic over Wi-Fi

Media import failed

Privacy Analysis of Augmented Reality Glasses

1 Introduction

This experiment captures network traffic over Wi-Fi during the interaction between an Android phone and the Ray Ban Meta glasses, more specifically while executing voice commands. There was an unexpected issue during capture where (i)Meta was not able to identify the person I wanted to communicate and send a message to and (ii)the phone was not able to import the photo/video from the glasses. This is similar to the case for 'Capturing traffic over Wi-Fi - Touch Commands - Side Capture - Media import failed'. Similarly, I continued capturing network traffic whilst troubleshooting to determine if any relevant data could be observed during the failure. The issue was resolved in the end by rebooting the glasses as well. The hope (goal) for keeping this data collection is to have a view of the glasses's behavior under error conditions, which providing insights into how the device handles errors and reboots.

2 Metadata

Time of experiment: Oct 16th 02:20

Duration of experiment: Roughly 50 minutes

IP addresses:

- Laptop - 192.168.0.215
- Redmi phone - 192.168.1.141
- Wi-Fi Network - 192.168.0.1
- ALFA Hotspot - 192.168.1.100/24
- Meta CDN (content delivery network) - 157.240.17.17

3 Voice Commands used

- "Hey Meta, what's the weather in [place] today?"
- "Hey Meta, what time is it?"
- "Hey Meta, what date is today?"
- "Hey Meta, what's the battery percentage?"

- "Hey Meta, play music."
- "Hey Meta, increase the volume."
- "Hey Meta, lower the volume."
- "Hey Meta, pause music."

- "Hey Meta, take a photo."

- "Hey Meta, take a video."
- "Hey Meta, stop recording."
- "Hey Meta, send a message to [name]."
- "Hey Meta, send a voice message to [name]."
- "Hey Meta, read my messages."
- "Hey Meta, set a timer for 1 minute."
- "Hey Meta, how much time is left in my timer."
- "Hey Meta, cancel my timer for 1 minute."

4 Time Stamps

The following is some timelines that should be noted.

The number is the packet number on Wireshark:

32701 reboot

33140 reboot done

35010 start app

38000 start import

41625 turn off glasses and close hinges turn on

48309 close app put glasses in case and reopened app

49110 turn off bluetooth

49494 turn on bluetooth then open app

52649 turn off glasses

52678 turn on glsases

54069 factory resetting glasses

67000 paired glasses

81948 connect to spotify tap

between other settings 86169 connect to instagram then left idle for 5 min

5 Note

Meta was not able to understand the name given to send the message to.

Meta could not import the media that ultimately resulted in a factory reset.